No

Date

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TUGAS 2 NGOTRET TEKKOM

Untuk 2 buah persamaan : $f(x) = x^2 - x - 2 = 0$ dan $g(x) = x^2 - 4 = 0$, hitunglah pendekatan utk nilai x sampai Iterasi ketiga metode regulafalsi.

Jawaban : Fungsi 1 : $f(x) = x^2 - x - 2 = 0$

Aturan Regulafalsi $\rightarrow$ $f(a) \cdot f(b) < 0$

Fungsi a dan b nilainya harus beda tanda

Maka : $a = 1 \rightarrow f(1) = 1^2 - 1 - 2 = -2$ (negatif)
$b = 3 \rightarrow f(3) = 3^2 - 3 - 2 = 4$ (positip) } Regulafalsi

$$\boxed{\text{Rumus Regulafalsi} : c = b - \frac{f(b)(b-a)}{f(b) - f(a)}}$$

Iterasi 1 :

$$c = 3 - \frac{4(3-1)}{4-(-2)} = 3 - \frac{8}{6} = 1{,}667 \rightarrow f(1{,}667) = -0{,}889$$

Interval yg dihasilkan $\rightarrow$ $a = 1{,}667$, $b = 3$

Iterasi 2 :

$$c = 3 - \frac{4(3-1{,}667)}{4-(-0{,}889)} = 1{,}909 \rightarrow f(1{,}909) = -0{,}265$$

Interval yg dihasilkan $\rightarrow$ $a = 1{,}909$, $b = 3$

Iterasi 3 :

$$c = 3 - \frac{4(3-1{,}909)}{4-(-0{,}265)} = 3 - \frac{4(1{,}091)}{4+0{,}265} = 3 - \frac{4{,}364}{4{,}265} = 3 - 1{,}023$$

$c = 1{,}977$

$a = 1{,}977$, $b = 3$

Dipindai dengan CamScanner

Jawaban : Fungsi 2 : $g(x) = x^2 - 4$

$a = 1$ > $g(1) = 1^2 - 4 = -3$ (negatif)
$b = 3$ $g(3) = 3^2 - 4 = 5$ (positif)

$$\boxed{c = b - \frac{f(b)(b-a)}{f(b) - f(a)}}$$

Iterasi 1 :

$$c = 3 - \frac{5(3-1)}{5-(-3)} = 3 - \frac{5(2)}{5+3} = 3 - \frac{10}{8} = 3 - 1{,}25 = 1{,}75$$

$a = 1{,}75$ $g(1{,}75) = -0{,}9375$
$b = 3$

Iterasi 2 :

$$c = 3 - \frac{5(3-1{,}75)}{5-(-0{,}9375)} = 3 - \frac{5(1{,}25)}{5{,}9375} = 3 - 1{,}052$$

$c = 1{,}947$

$a = 1{,}947$ $g(1{,}947) = -0{,}21$
$b = 3$

Iterasi 3 :

$$c = 3 - \frac{5(3-1{,}947)}{5-(-0{,}21)} = 3 - \frac{5(1{,}053)}{5{,}21} = 3 - \frac{5{,}265}{5{,}21} = 3 - 1{,}0105$$

$c = 1{,}990$

Dipindai dengan CamScanner